

AI-Powered Customer Churn Prediction in the Telecom Industry

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1. Introduction

This project aims to address the critical business problem of customer churn in the telecommunications industry. Customer churn, the rate at which customers discontinue their service, is a major concern for companies as acquiring new customers is significantly more expensive than retaining existing ones. The motivation behind this project is to leverage machine learning to build a predictive model that can accurately identify customers who are likely to churn. By understanding the key drivers of churn and predicting at-risk customers, a telecom company can proactively implement targeted retention strategies, such as special offers or improved customer service, to reduce revenue loss and improve customer loyalty.

2. Dataset Description

Dataset Overview

The project utilizes the "Telco Customer Churn" dataset, which contains customer data and their churn status.

1. **How many features?** The dataset initially contains 21 features (columns). After removing customerID, 20 features are used for modeling.
2. **How many data points?** The dataset contains 7043 data points (rows), each representing a unique customer.
3. **Classification or regression problem?** This is a **classification problem**. The goal is to predict a categorical outcome: whether a customer will **Churn ('Yes')** or **Not Churn ('No')**. The target variable is binary, not a continuous value.

Feature Analysis

1. **What kind of features are in your dataset?** The dataset contains a mix of feature types:
    a. **Quantitative (Numerical) Features**: tenure, MonthlyCharges, TotalCharges. These represent measurable quantities.
    b. **Categorical Features**: gender, SeniorCitizen, Partner, Dependents, PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies, Contract, PaperlessBilling, PaymentMethod. These represent qualitative characteristics or groups.
2. **Do you need to encode the categorical variables? Why?** Yes, encoding categorical variables is essential. Machine learning algorithms, including Logistic Regression and Neural Networks, are

based on mathematical equations and require numerical input. They cannot process text-based categories like 'DSL' or 'Month-to-month' directly. Therefore, techniques like One-Hot Encoding and Label Encoding were used to convert these text labels into a numerical format that the models can understand.

Correlation Analysis

A heatmap was generated to visualize the correlation between all numerical features, including the encoded target variable 'Churn'.

1. **What do you understand after the correlation test?** The correlation heatmap reveals several key relationships:
    a. **Positive Correlation with Churn**: MonthlyCharges has a noticeable positive correlation with Churn. This suggests that customers with higher monthly bills are more likely to leave.
    b. **Negative Correlation with Churn**: tenure and Contract (specifically longer-term contracts after encoding) show a strong negative correlation with Churn. This is intuitive: long-term, loyal customers are far less likely to churn.
    c. **Multicollinearity**: TotalCharges is highly correlated with tenure and MonthlyCharges. This is expected, as total charges are a function of the monthly rate and the duration of service.

3. Imbalanced Dataset Analysis

1. **For the output feature, do all unique classes have an equal number of instances?** No, the classes in the output feature 'Churn' are not balanced. There is a significant majority of customers who did not churn compared to those who did.
2. **Represent using a bar chart:** A bar chart of the 'Churn' variable shows this imbalance clearly. Out of 7043 customers, over 5000 have not churned, while only around 1800 have churned. This indicates that the dataset is imbalanced, which can bias a model towards predicting the majority class. This was addressed by using stratified splitting.

4. Exploratory Data Analysis (EDA)

EDA was performed to uncover patterns related to customer churn:

1. **Churn by Contract Type**: Customers on a **'Month-to-month'** contract have a dramatically higher churn rate compared to those on 'One year' or 'Two year' contracts. This is the single most significant indicator of churn risk.
2. **Churn by Tenure**: Churn is heavily concentrated among customers with very low tenure (0-12 months). As tenure increases, the likelihood of churning decreases significantly.
3. **Churn by Internet Service**: Customers with **'Fiber optic'** internet service have a higher churn rate than those with 'DSL', possibly due to higher costs or service issues associated with that plan.

5. Dataset Pre-processing

Several data quality issues were identified and addressed to prepare the dataset for modeling.

1. **Fault: Null / Missing Values** Initially, df.isnull().sum() showed no missing values. However, the TotalCharges column was of type 'object' because it contained empty spaces (' ') for new customers with 0 tenure. When this column was converted to a numeric type, these spaces became NaN (Not a Number) values.
    a. **Solution: Imputation**: The NaN values in TotalCharges were imputed using the **median** of the column. The median was chosen over the mean because it is more robust to outliers. This imputation was performed after the train-test split to prevent data leakage from the test set into the training set.
2. **Fault: Categorical Values** The dataset contained numerous text-based categorical features that needed to be converted to a numerical format.
    a. **Solution: Encoding**: A combination of encoding strategies was used. **Dummy Encoding (One-Hot Encoding)** was applied to nominal features with multiple categories (e.g., Contract, PaymentMethod) to create binary columns for each category. **Label Encoding** was used for binary categorical features (e.g., gender, Partner) to convert them to 0s and 1s.
3. **Fault: Feature Scaling and Skewness** The numerical features (tenure, MonthlyCharges, TotalCharges) had different scales and ranges. Additionally, TotalCharges was found to be right-skewed.
    a. **Solution: Scaling and Transformation**: To address the different scales, StandardScaler was used on all numerical features. To correct the skewness in TotalCharges, a **Power Transformer** (yeo-johnson method) was applied, making its distribution more normal and helping models that assume a Gaussian distribution.

6. Dataset Splitting

The pre-processed dataset was split into a training set and a testing set.

1. **Method**: A **stratified split** was used.
2. **Ratio**: The data was split into **70% for the training set** and **30% for the test set**.
3. **Justification**: Stratification was essential due to the imbalanced nature of the 'Churn' target variable. It ensures that both the training and test sets have the same proportion of 'Yes' and 'No' churn instances as the original dataset, leading to a more reliable model evaluation.

7. Model Training & Testing (Supervised Learning)

Three different classification models were trained and evaluated.

Model 1: Logistic Regression

A Logistic Regression model was trained. The max_iter parameter was increased to 5000 to ensure the model had sufficient iterations to converge.

Model 2: Decision Tree Classifier

A Decision Tree was trained. To prevent overfitting and improve stability, the model's complexity was constrained by setting max_depth=5 and min_samples_leaf=10. A random_state was set for reproducibility.

Model 3: Neural Network (MLP Classifier)

A Multi-layer Perceptron (Neural Network) was implemented. The architecture was manually defined with two hidden layers containing 30 and 30 neurons, respectively. The 'relu' activation function was used for the hidden layers, and the 'adam' solver was used for optimization with a learning rate of 0.001.

8. Unsupervised Learning: K-Means Clustering

1. To treat the problem as an unsupervised learning task, K-Means clustering was applied to segment customers into distinct groups.
2. **Process**: The **Elbow Method** was first used to determine the optimal number of clusters, which was found to be **3**. The K-Means algorithm was then run with k=3 on the preprocessed training data.
3. **Showcasing the Clusters**: The analysis revealed three distinct customer segments:
    a. **Cluster 0**: A high-churn, high-risk segment characterized by very low tenure and high monthly charges.
    b. **Cluster 1**: A loyal, low-risk segment with high tenure and varied monthly charges.
    c. **Cluster 2**: A moderate-risk segment with average tenure and lower monthly charges.
4. **Insight**: This unsupervised approach successfully identified a high-risk customer profile without using the churn labels, demonstrating that distinct behavioral groups exist within the customer base.

9. Model Selection & Comparison Analysis

The performance of the three supervised models was compared using several metrics.

1. **Accuracy Comparison**: A bar chart showed that all models performed similarly, with accuracies around 80%. Logistic Regression and the Neural Network slightly outperformed the Decision Tree.
    a. **Logistic Regression Accuracy**: 80.73%
    b. **Decision Tree Accuracy**: 79.08%
    c. **Neural Network Accuracy**: 80.40%
2. **Confusion Matrices**: Confusion matrices were generated for each model, providing a detailed breakdown of their predictions. The Logistic Regression model had the highest number of True Negatives (1412), indicating it was very good at correctly identifying customers who would not churn.
3. **ROC Curve and AUC Score Comparison**: An ROC curve plot was generated to compare the models' ability to distinguish between the two classes.
    a. **Logistic Regression**: AUC = 0.85
    b. **Decision Tree**: AUC = 0.82
    c. **Neural Network**: AUC = 0.85

The Logistic Regression model and Neural Network had the highest Area Under the Curve (AUC), indicating the best overall classification performance.

4. **Precision and Recall Comparison**: The Logistic Regression model achieved the best balance of precision (68%) and recall (52%) for the positive 'Churn' class among the models tested.

10. Conclusion

1. **Understanding the Results**: The analysis successfully identified key drivers of churn, with contract type, tenure, and internet service being the most influential factors. All three supervised models demonstrated strong predictive power, achieving accuracies of approximately 80%. The **Logistic Regression model emerged as the top performer**, with the highest AUC score (0.85) and a good balance of precision and recall.
2. **Model Performance Comments**: The high accuracy of the models is encouraging, but the class imbalance means we must also consider precision and recall. The Logistic Regression model's ability to correctly identify over 80% of actual non-churners (high specificity) while still catching 55% of actual churners (recall) makes it a valuable and reliable tool for the business. The K-Means clustering analysis complemented this by identifying a specific high-risk customer persona that retention efforts could be focused on.
3. **Reasons for Results**: Logistic Regression likely performed best because the relationships in the data (e.g., lower tenure leads to higher churn) are relatively linear and can be captured well by this model. The constrained Decision Tree was simpler and slightly less powerful, while the Neural Network, despite its complexity, did not significantly outperform the simpler logistic model, suggesting that a more complex architecture was not necessary for this dataset.
4. **Challenges Faced**: The primary challenge was the subtle data quality issue in the TotalCharges column, where empty strings initially masked the presence of missing values. Identifying and correctly imputing these values after the train-test split was a critical step. Another challenge was interpreting the models in the context of the imbalanced dataset, which required looking beyond simple accuracy to metrics like AUC, precision, and recall.